

Total No. of Questions : 6]

SEAT No. :

**P208**

[Total No. of Pages : 2

**APR - 17/TE/Insem. - 44**

**T.E. (Information Technology)**

**DESIGN & ANALYSIS OF ALGORITHMS**

**(2012 Pattern) (Semester - II)**

*Time : 1 Hour]*

*[Max. Marks : 30*

*Instructions to the candidates:*

- 1) Answer Q.1 or Q.2, Q.3 or Q.4 Q.5 or Q.6.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) Set up a recurrence relation to compute  $n!$  and solve it. [5]
- b) Prove by method of contradiction that “There is no greatest even interger”. [5]

OR

- Q2)** a) Discuss the general plan for analyzing efficiency of non-recursive algorithm. [5]
- b) Prove by method of contraposition that  $x + 8$  is odd. [5]

- Q3)** a) Write an algorithm for binary search and find the worst case efficiency. [5]
- b) Show the steps in multiplying the following two intergers using efficiency integer multiplication method:  $2101 * 1130$ . [5]

OR

- Q4)** a) Construct Huffman tree for the following data and obtain it's Huffman's code. [5]

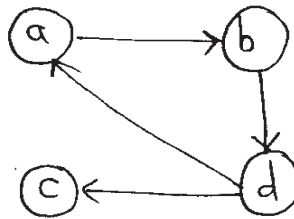
|             |     |      |     |     |     |     |
|-------------|-----|------|-----|-----|-----|-----|
| Character   | A   | B    | C   | D   | E   | -   |
| Probability | 0.5 | 0.35 | 0.5 | 0.1 | 0.4 | 0.2 |

*P.T.O.*

- b) Apply Strassen's algorithm to compute: [5]

$$\begin{bmatrix} 1 & 2 & 1 & 1 \\ 0 & 3 & 2 & 4 \\ 0 & 1 & 1 & 1 \\ 5 & 0 & 1 & 0 \end{bmatrix} * \begin{bmatrix} 2 & 1 & 0 & 2 \\ 1 & 2 & 1 & 1 \\ 0 & 3 & 2 & 1 \\ 4 & 0 & 0 & 4 \end{bmatrix}$$

- Q5) a) Use Warshall's method to find transitive closure. [5]



- b) Solve the following instance of knapsack problem using dynamic programming. [5]

| Item   | 1    | 2    | 3    | 4    |
|--------|------|------|------|------|
| Weight | 2    | 1    | 3    | 2    |
| Value  | \$12 | \$10 | \$20 | \$15 |

OR

- Q6) a) Construct an optimal binary search tree for data  $n = 4$ ,  $(a_1, a_2, a_3, a_4) = (\text{do}, \text{if}, \text{int}, \text{while})$ ,  $p(1:4) = (3, 3, 1, 1)$  and  $q(0:4) = (2, 3, 1, 1, 1)$  [6]
- b) Write Flyod's algorithm to find all pairs shortest paths problem. [4]

